

Report of Calibration
PSC 2CH-HSS-AR-QD-QF_S/N 0004
2025-11-24 14:25:55

Calibration Current standard: BNL PSC ATE S/N 001
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647
Calibration Resistance standard: Fluke 742A-1 S/N 1063008
End Header

lab{5}Chan1
Burden resistor = 18.0000

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000613	0.998095	-1.028164	-1.026721	1.000979	0.038635
-50.005970	50.024637	-49.663334	-49.657726	50.123871	0.062210

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.002951	-0.035109	-0.033752	0.000000
Initial measured gains:	1.000432	0.992446	0.992361	1.001965
Gain corrections:	1.000432	1.007611	1.007698	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.000633
50.000000	50.118137

Measured offset: -0.001765
Measured gain: 1.002398
Gain correction: 0.997608

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000575	1.000312	-1.000626	-1.000731	1.000907	0.036731
-50.005957	50.005201	-50.005547	-50.005535	50.006420	0.055118

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000252	-0.000060	-0.000168	0.000582
Final measured gains:	0.999990	0.999991	0.999988	1.000013

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000088	-0.000065	-0.000058	0.000128
Final measured offsets stdev:	0.000253	0.000156	0.000195	0.000571
Final measured gains mean:	0.999985	0.999984	0.999986	1.000000
Final measured gains stdev:	0.000013	0.000007	0.000007	0.000009

Saving channel 1 calibration constants to qspi

lab{5}Chan2
Burden resistor = 9.0000

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000274	0.991256	-1.060808	-1.063322	1.002166	0.017996
-100.012127	100.034771	-99.313034	-99.321022	100.240738	0.133546

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.009338	-0.068208	-0.070667	0.000000
Initial measured gains:	1.000320	0.992328	0.992383	1.001969
Gain corrections:	1.000320	1.007731	1.007675	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.001476
100.000000	100.228233

Measured offset: -0.000814
Measured gain: 1.002290
Gain correction: 0.997715

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000176	1.000203	-1.000406	-1.000319	1.001578	0.022364
-100.012208	100.011140	-100.011261	-100.011177	100.012825	0.133804

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000038	-0.000242	-0.000154	0.001371
Final measured gains:	0.999989	0.999988	0.999988	1.000003

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000148	-0.000235	-0.000090	0.001594
Final measured offsets stdev:	0.000279	0.000079	0.000155	0.000175
Final measured gains mean:	0.999985	0.999986	0.999988	0.999987
Final measured gains stdev:	0.000009	0.000005	0.000006	0.000010

Saving channel 2 calibration constants to qspi

Test data reviewed by  Date 11/25/25